

Plant Disease Detection Using CNN and Automated Chatbot

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Abstract— Plant diseases are a significant threat to global agriculture, leading to reduced crop yields and economic losses. Early detection and timely intervention are crucial to mitigate these effects. Using Convolutional Neural Networks (CNN) in conjunction with a chatbot for real-time illness diagnosis and remedy recommendation, this paper suggests a unique method for plant disease detection. While the chatbot offers farmers practical answers, the CNN model is trained on a dataset comprising images of healthy and diseased plant leaves to precisely diagnose ailments. Unlike pre-trained models like VGG19 or EfficientNet, which demand high computation and fine-tuning, the proposed custom CNN model is scalable, with experimental results confirming its efficiency. Farmers with little to no technical knowledge can utilize the system because of the automated Google Gemini chatbot and the user-friendly interface developed using streamlit. By fusing deep learning with sophisticated language models for sustainable farming methods, this work advances the field of precision agriculture.

Keywords— *Plant Disease Detection, Convolutional Neural Network (CNN), Chatbot, Precision Agriculture, Google Gemini, Streamlit.*

I. INTRODUCTION

The rapid growth of digital platforms has changed the way how information is shared, especially in agriculture. Farmers are increasingly turning to digital tools for managing crops and diagnosing diseases. However, identifying plant diseases in a timely manner is still quite difficult. Delays in detecting infections can lead to substantial crop losses, lower yields, and financial instability. Traditional methods of manual inspection are not only time-consuming and labor-intensive but also susceptible to errors and inefficiencies, particularly in large-scale farming operations. As deep learning and artificial intelligence (AI) advance, there is now a significant opportunity to automate and enhance the precision of plant disease detection. Convolutional Neural Networks (CNNs) have demonstrated remarkable efficacy in categorization

tasks using images, demonstrating exceptional proficiency in identifying patterns and deriving significant features from visual data. However, while only CNNs perform well in identifying diseases, they do not offer practical recommendations for farmers, creating a gap in the decision-making process. This limitation highlights the necessity for a comprehensive system that not only detects plant diseases accurately but also provides real-time guidance for effective disease management.

Chatbots powered by Google Gemini's Natural Language Processing (NLP) capabilities provide instant and personalized responses, making them an excellent addition to CNN-based disease detection systems. By combining these technologies, the proposed system improves plant disease management by delivering both accurate diagnoses and actionable treatment recommendations. This paper presents a hybrid framework that combines an intelligent chatbot for real-time disease management with the advantages of CNNs for spatial feature extraction. To provide accurate and trustworthy identification, the suggested method employs a Convolutional Neural Network (CNN) model that has been designed specially for plant disease detection. The integrated chatbot, powered by Google Gemini and accessible via an interactive Streamlit UI, enables smooth interaction with farmers, offering effective recommendations for disease management.

II. CONTRIBUTIONS

1. Developed a custom CNN model using TensorFlow for accurate plant disease classification.
2. Integrated a chatbot powered by Google Gemini, which automatically receives the model's predictions to provide instant remedy recommendations.
3. Designed an interactive Streamlit UI for seamless image uploads, disease predictions, and chatbot interactions.
4. Evaluated the model accuracy and usability on real-world datasets.

III. RELATED WORK

Notable progress has been made recently thanks to developments in deep learning, and computer vision technology, leading to advancements in the detection and categorization of plant diseases. The scalability and accuracy of early methods, which depended on color-based segmentation and texture analysis, were constrained by differences in lighting, leaf orientation, and disease symptoms. The extensive usage of Convolutional Neural Networks (CNNs) for the detection of plant diseases is a result of the development of deep learning. Mohanty et al. showcased cutting-edge accuracy by employing CNNs on the PlantVillage dataset, removing the necessity for manual feature extraction. Similarly, Sladojevic et al. created a CNN-based framework that delivered high precision in identifying crop diseases.

1. CNNs with chatbot:

Despite their effectiveness, many CNN-based systems focus exclusively on disease detection without providing practical solutions. Researchers have delved into the use of chatbots to address this problem. Patil et al. proposed a chatbot-driven advisory system that offered real-time crop management advice through NLP-based query interpretation. Hybrid approaches, such as the CNN-LSTM model developed by Chen et al., have further improved detection accuracy by integrating spatial and sequential data analysis. The development of user-friendly interfaces and real-time solutions has also gained traction. Kumar et al. designed a mobile application that combines CNNs with a chatbot, allowing farmers to upload images, obtain diagnoses, and access treatment recommendations. However, challenges such as dataset limitations, model generalization, and multilingual support still need to be addressed.

2. VGG19 model:

Detection and classification of plant diseases is one of the many picture classification jobs that make considerable use of the VGG19 model, a very sophisticated convolutional neural network (CNN). With astonishing accuracy, it analyzes leaf photos to identify disease patterns. Its large number of parameters, however, raises computing costs and makes real-time implementation challenging, especially in resource-constrained contexts. Moreover, VGG19 requires significant data preprocessing and augmentation to manage variations in lighting, angles, as well as environmental elements, which might affect how well it performs.

VGG19 excels at capturing intricate visual patterns and extracting detailed features, enabling precise classification. As a pretrained model, it supports transfer learning, reducing the time required for training. Nevertheless, it demands substantial computational power and relies on high-quality, diverse datasets to achieve effective generalization. Training and fine-tuning the model require expertise in deep learning, and its "black box" nature limits its interpretability. Additionally, distinguishing between illnesses that are similar visual symptoms and adapting to environmental changes

remain ongoing challenges. Plant diseases pose a major threat to the world's food security since they have a significant effect on crop output and health. Timely and precise disease detection is essential for effective crop management.

3. EfficientNET model:

EfficientNet, a sophisticated deep learning model, tackles the issue of class imbalance often found in plant disease datasets. This model improves the diagnosis of uncommon illnesses that are usually missed by traditional techniques by using a varied collection of plant photos. Focal Loss increases the model's sensitivity to subtle disease symptoms, which are often missed by standard cross-entropy loss models. EfficientNet enhances the identification of rare diseases by effectively addressing class imbalance. Its architecture is optimized to use minimal computational resources while providing excellent performance. The model is particularly adept at detecting mild or visually similar disease symptoms, improving overall classification accuracy. However, it may still struggle with diseases that have overlapping or subtle symptoms. The effectiveness of the model depends heavily on well-labeled, high-quality datasets, emphasizing the importance of thorough data preparation. Distinguishing between similar illnesses remains a challenge, and variations in lighting, leaf orientation, and background can impact accuracy. Despite these advancements, occasional misclassifications persist, indicating the need for further refinement of the model.

IV. PROBLEM STATEMENT

Plant diseases present a major challenge to global food security, causing significant declines in crop yield and quality. Conventional techniques for illness detection rely on manual inspection, which is time-consuming, labor-intensive, and prone to human error. In order to decrease crop losses and enhance agricultural productivity, an automated and accurate detection system is crucial. The goal of this project is to create a plant disease detection model/system that analyzes pictures of leaves using Convolutional Neural Networks (CNNs). The suggested approach seeks to offer a quicker solution for farmers through the automated chatbot and agricultural stakeholders, facilitating early detection and effective disease management.

V. PROPOSED WORK

A deep learning-based framework has been made for the identification and categorization of plant diseases, aiming to boost agricultural productivity. This approach combines EfficientNet for feature extraction using a neural network that is convolutional to precisely classify plant diseases. Furthermore, an AI-powered chatbot offers real-time recommendations and solutions for disease management to farmers. The dataset, obtained from PlantVillage, includes images of various crops and their disease conditions. By automating the disease identification process, the system improves accuracy while reducing the requirement for personal intervention. The proposed chatbot framework delivers high classification accuracy and surpasses traditional

machine learning methods. The primary objective is to create a user-friendly system for farmers. Key objectives include:

- Classify plant diseases using CNNs.
- Provide real-time solutions via chatbot.
- Design an intuitive interface for ease of use

1. Methodology:

The plant disease detection process follows a structured approach, beginning with data preprocessing. This step involves cleaning the dataset by resizing, normalizing, and augmenting plant leaf images to ensure consistency and improve model performance. After then, the pictures are converted into a numerical format appropriate for deep learning. A specially constructed Convolutional Neural Network (CNN) is used by the system for effective and precise classification. The unique CNN architecture consists of convolutional layers to extract features, batch normalization for training stability, ReLU activation to prevent vanishing gradients, max-pooling for dimensionality reduction, dropout layers to avoid overfitting, and final categorization using fully connected layers. This design ensures a balance between depth and computational efficiency, making it highly effective for large-scale plant disease detection. Compared to VGG-19, which has around 143M parameters and slower inference speed, the custom CNN is lightweight (~5M parameters), significantly faster, and requires less computational power without compromising accuracy. While EfficientNet offers better performance for imbalanced datasets, it remains computationally heavier and slower than the custom model.

The custom CNN was selected due to their superior trade-off between accuracy, inference speed, and resource efficiency, guaranteeing the efficient identification. The project's findings show that the CNN-based paradigm, combined with the chatbot, achieves high accuracy and effectively handles class imbalances, providing reliable disease detection and actionable solutions for farmers.

2. Dataset:

The “PlantVillage” dataset is a well-liked option for plant disease diagnosis, and it is sourced from Kaggle. It includes labeled images of various crops (e.g., tomato, grape, apple) and their disease conditions in colour, grayscale, and segmented formats.

The preprocessing procedures for identifying plant diseases ensure a high-quality dataset and improve model performance. Data cleaning entails getting rid of damaged and duplicate pictures. Resizing and normalization standardize images to 224×224 pixels and normalize pixel values to the [0,1] range.

3. System Implementation and Evaluation Methodology:

The evaluation of the suggested system for detection of plant diseases was conducted using a systematic methodology that

combined model development, dataset preparation, and consistent input. Data augmentation techniques such as random rotations, zooming, flipping, and brightness adjustments prevent overfitting and improve model generalization. The dataset is then split into 80% training and 20% validation to ensure effective learning and evaluation. To optimize preprocessing, TensorFlow/Keras image data generators enable real-time augmentation during training, enhancing model robustness.

4. Algorithm:

Step 1: Data Preprocessing: Resize, normalize, and augment plant leaf images for consistency and improved model performance.

Step 2: Feature Extraction: Use a custom-built CNN with batch normalization, ReLU, convolutional layers, max-pooling, and dropout for efficient feature extraction.

Step 3: Model Training: Train the custom CNN on the preprocessed dataset for accurate plant disease classification.

Step 4: Chatbot Integration: Implement a chatbot using the Google Gemini API key to provide real-time disease solutions based on CNN predictions.

Step 5: Model Evaluation: Assess the CNN model using recall, accuracy, precision, and F1-score.

Step 6: System Deployment: Implement the integrated CNN and chatbot system on a Streamlit-based interface.

5. Dataset Description:

S. No.	Attributes	Description
1.	Image ID	An individual identification for every picture
2.	Plant Species	The type of plant (e.g., tomato, apple, soybean, strawberry).
3.	Disease Label	The disease affecting the plant or "healthy" if no disease is present.
4.	Image Type	Colour, grayscale, or segmented versions of the image.
5.	Image Format	Common formats include JPG and PNG.
6.	Image Size	Varies but typically resized to 224×224 pixels to train model.
7.	Responses	Chatbot-generated responses to the relevant queries.

Table 1. Dataset Description

6. Convolution Neural Network (CNN):

CNNs capacity to identify spatial patterns and extract significant characteristics from leaf photos makes them useful for detecting plant diseases. Two convolutional layers and max pooling layers make up the model architecture, which facilitates effective feature extraction. Before being delivered to the output layer, which has 38 neurons, each of which represents a distinct illness or healthy category, the collected characteristics are processed by a fully linked dense layer with 256 neurons.

7. Adam Optimizer:

In order to provide steady convergence and effective weight updates during training, Adam optimizer is chosen to dynamically modify the learning rate. This optimization method is appropriate for managing intricate datasets with a high number of parameters because it strikes a balance between exploration and exploitation.

8. Categorical Cross-Entropy Loss:

The difference between expected probability and actual class labels is measured using categorical cross-entropy loss since the dataset contains several classes. The model learns clear class separations during several training epochs with the aid of this loss function, which penalizes inaccurate predictions.

9. Activation Function:

The hidden layers employ the Rectified Linear Unit (ReLU) activation function to add non-linearity and boost learning effectiveness. To manage multi-class classification, the output layer applies the softmax activation function, which guarantees that the highest probability prediction matches the appropriate illness class.

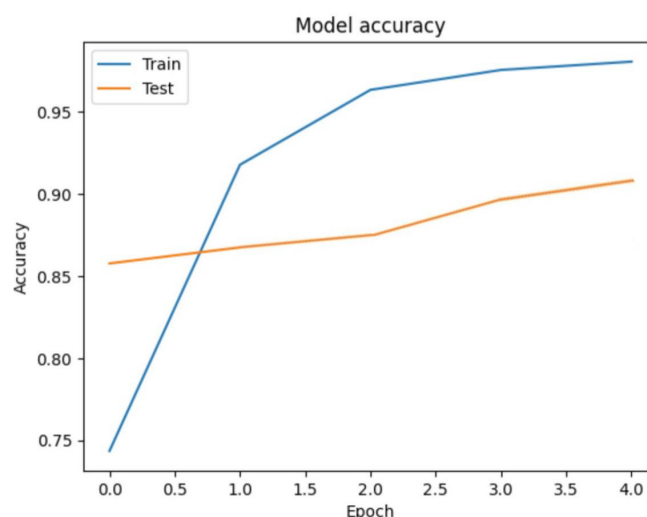
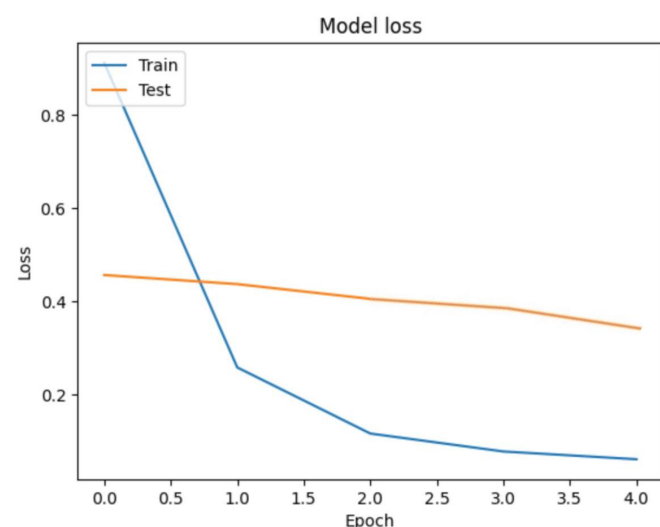


Fig 1. Model Loss and Accuracy Graph

10. Integrated Chatbot:

The chatbot is developed using the Google Gemini API to offer real-time interaction and personalized disease management recommendations. It allows farmers to upload images of plant leaves, receive instant disease diagnoses, and ask relevant questions regarding crop health. By providing immediate responses and actionable insights, the chatbot enhances the accessibility and user-friendliness of the system. Providing farmers with relevant knowledge to enable them for effective decision-making.

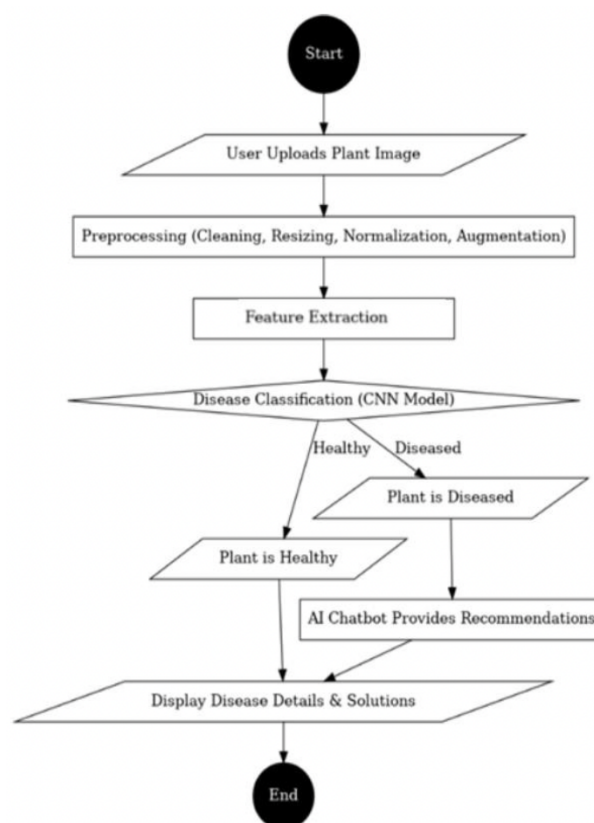


Fig 2. Flow Chart

VI. RESULTS AND DISCUSSION

Plant Disease Detection: Achieved an exceptional accuracy of 90%, ensuring highly precise identification of plant illnesses from pictures of leaves. This enhances early detection, helping farmers take timely action to prevent crop damage.

Chatbot Integration: Implemented an advanced AI- powered chatbot using Google Gemini API, providing users with instant disease diagnosis and treatment recommendations. This interactive system improves accessibility and decision-making for effective plant health management.

The below table presents the epoch-wise performance metrics of the custom CNN model educated to recognize plant diseases over five epochs. It tracks key indicators such as accuracy, loss, validation accuracy, validation loss, and training time per epoch.

Epochs	Accuracy	Loss	Validation Accuracy	Validation Loss	Time per Epoch (s)
1	0.6046	1.6191	0.8577	0.4560	122
2	0.9184	0.2622	0.8674	0.4365	140
3	0.9669	0.1100	0.8721	0.4049	104
4	0.9791	0.0675	0.8923	0.3939	147
5	0.9822	0.0542	0.9000	0.3517	137

Table 2: Epoch wise performance metrics

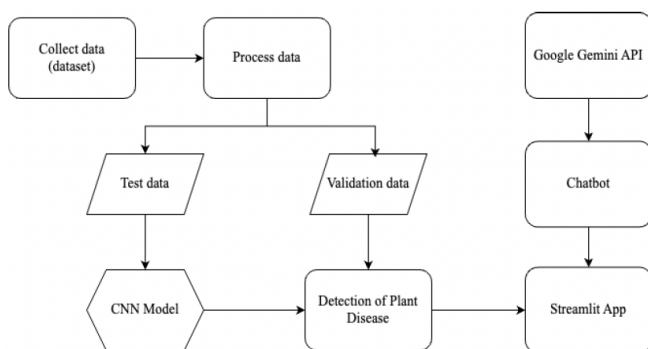


Fig 3: Model

Performance Metrics:

1. Accuracy:

Measures how often the model predicts correctly.

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$

Use: Overall performance check.

2. Precision:

Predicts the count of actual positives.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

Use: Important when the cost of false positives is high.

3. Recall:

How many actual positives are correctly predicted.

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

Use: Critical when missing a positive case is costly.

4. F1 Score:

Precision and recall harmonic mean

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Use: Beneficial when recall and precision must be balanced

VII. CONCLUSION

In conclusion, the suggested plant disease system, combined with a chatbot for real-time solution recommendations, constitutes a noteworthy breakthrough in precision agriculture. By leveraging CNNs for accurate disease identification and using a chatbot for interactive user engagement, the system addresses critical agricultural challenges, including timely disease detection, accessibility, and decision-making support. This integration ensures that farmers receive precise disease diagnoses and also obtain actionable recommendations for effective disease management.

The CNN-based model's great accuracy in identifying plant illnesses under a variety of environmental situations is confirmed by experimental tests. Instant, context-aware replies are made possible by the chatbot, which is powered by Google Gemini and greatly improves system use. Improved disease detection accuracy, real-time user engagement for

smooth communication, and effective processing of big datasets with little computing cost are some of the main results of the suggested approach.

The proposed system addresses the limitations of existing models by combining a custom CNN with a chatbot-driven advisory system, ensuring both high accuracy (90%) in plant disease classification and real-time remedy recommendations. Future enhancements could include expanding the dataset to cover a broader range of plant species and disease categories, optimizing the model for real-time deployment on mobile devices, and incorporating multilingual support to cater to a diverse user base.

These advancements will further improve the system's efficiency and usability, making it a robust tool for precision agriculture. Overall, this research underscores the potential of AI-driven solutions in enhancing crop health, improving agricultural productivity, and promoting sustainable farming practices. The incorporation of deep learning and chatbot-based interaction in plant disease detection establishes a strong foundation for future advancements in smart agriculture.

VIII. REFERENCES

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